BEING PROPER OVER THE BASE DESCENDS ALONG AN FPQC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the notion of being proper over the base by a label in the cohomology-of-schemes chapter; the reference is carried as an external reference stub, and the definition is not reproduced.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The reduced induced closed subscheme structure on a closed subset. Schemes, Definition schemes-definition-reduced-induced-scheme, invoked at proof lines 6737-6738. Turns the closed subset into a scheme so that properness of a morphism can be discussed.
- Being proper over the base, and its characterisation through closed subschemes. Cohomology of Schemes, Definition coherent-definition-proper-over-base at statement lines 6731-6732 and Cohomology of Schemes, Lemma coherent-lemma-closed-proper-over-base at proof lines 6741-6743. Supplies the definition being descended and the step from the closed subset condition to properness of the projections.
- Properness descends along an fpqc covering. Same chapter, Lemma lemma-descending-property-proper, invoked at proof lines 6744-6745. Gives properness of the morphism from the reduced induced scheme to the base.

Source reference. Original definition 0CYM: sources/stacks-chow/tags.txt, source lines 16247–16247. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CYL: sources/stacks-chow/tags.txt, source lines 16246–16246. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L1: sources/stacks-descent/pinned-original-source.tex, source lines 6037–6037. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes which is locally of finite type. Let Z be a closed subset of X. If there exists an fpqc covering $\{Y_i \to Y\}$ such that the inverse image $Z_i \subset Y_i \times_Y X$ is proper over Y_i (Cohomology of Schemes, Definition 0CYM) then Z is proper over Y.*

Proof. Endow Z with the reduced induced closed subscheme structure, see Schemes, Definition 01J4. For every i the base change $Y_i \times_Y Z$ is a closed subscheme of $Y_i \times_Y X$ whose underlying closed subset is Z_i. By definition (via Cohomology of Schemes, Lemma 0CYL) we conclude that the projections $Y_i \times_Y Z \to Y_i$ are proper morphisms. Hence $Z \to Y$ is a proper morphism by Lemma 02L1. Thus Z is proper over Y by definition. □